

Dynamic Guideline: WHO-Based Tuberculosis - Diagnosis and Treatment and Management

I. WHO Latest Guideline Summary (Summary Date: 29 April 2026)

1.1 Guideline Overview

Five Key WHO Guidelines Synthesized:

1. WHO consolidated guidelines on tuberculosis. Module 4: treatment - drug-resistant tuberculosis treatment, 2022 update

- Publication Date: June 2022
- Scope: Comprehensive guidance on treatment regimens for multidrug-resistant and rifampicin-resistant TB (MDR/RR-TB), including shorter all-oral regimens.

2. WHO consolidated guidelines on tuberculosis. Module 3: diagnosis - rapid diagnostics for tuberculosis detection, 2024 update

- Publication Date: 2024
- Scope: Recommendations on molecular WHO-recommended rapid diagnostic tests (mWRDs) including Xpert MTB/RIF Ultra, Truenat, and targeted next-generation sequencing (tNGS) for TB detection and drug-resistance profiling.

3. WHO consolidated guidelines on tuberculosis. Module 1: prevention - tuberculosis preventive treatment, 2nd edition

- Publication Date: 2024
- Scope: Updated guidance on TB preventive treatment (TPT) for people at risk, including people living with HIV, household contacts, and other at-risk populations. Includes 3HP (weekly rifapentine plus isoniazid for 3 months) and other short-course regimens.

4. WHO consolidated guidelines on tuberculosis. Module 6: tuberculosis and comorbidities, 2nd edition

- Publication Date: 2024
- Scope: Management of TB in patients with comorbidities including HIV, diabetes mellitus, mental health disorders, and substance use disorders. Addresses integrated care delivery models.

5. Meaningful youth engagement in a WHO guideline development process: case study of WHO tuberculosis guideline

- Publication Date: 29 April 2026
- Scope: Documents the process and outcomes of integrating youth perspectives into WHO TB guideline development, with implications for patient-centred care

recommendations and community engagement strategies.

Most Recent Guideline Details:

Field	Detail
Title	Meaningful youth engagement in a WHO guideline development process: case study of WHO tuberculosis guideline
Publication Date	29 April 2026
Target Population	Young people affected by TB (ages 10–24), guideline developers, national TB programme managers
Scope and Objectives	To describe and evaluate the systematic engagement of youth in WHO TB guideline development; to provide a replicable framework for meaningful youth participation in health policy; to ensure TB guidelines reflect the lived experiences and priorities of young people

1.2 Key Recommendations

A. Diagnosis

Recommendation	GRADE Evidence Quality	Strength
Xpert MTB/RIF Ultra should be used as the initial diagnostic test for TB in adults and children with signs and symptoms of pulmonary TB	High	Strong
Truenat MTB and MTB Plus assays may be used as initial tests for the diagnosis of TB and detection of rifampicin resistance	Moderate	Conditional
Targeted next-generation sequencing (tNGS) should be used for detection of resistance to isoniazid, fluoroquinolones, and second-line injectable agents following a positive molecular test	Moderate	Conditional
Lateral flow urine lipoarabinomannan (LF-LAM) assay should be used to assist in the diagnosis of active TB in people living with HIV with signs and symptoms of TB, or who are seriously ill, or with CD4 count ≤ 200 cells/ μ L	Moderate	Strong
Stool-based molecular testing (Xpert MTB/RIF Ultra on stool) may be used as an initial diagnostic test in children with signs and symptoms of pulmonary TB	Low	Conditional
Computer-aided detection (CAD) software may be used in place of human readers for interpreting digital chest X-rays for TB screening and triage	Low	Conditional

B. Treatment — Drug-Susceptible TB

Recommendation	GRADE Evidence Quality	Strength
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New patients with pulmonary TB should receive a 6-month regimen: 2 months of isoniazid, rifampicin, pyrazinamide, and ethambutol (HRZE) followed by 4 months of isoniazid and rifampicin (HR)	High	Strong
Fixed-dose combination (FDC) tablets are recommended over separate drug formulations	Moderate	Conditional
Daily dosing throughout treatment is recommended over intermittent dosing	High	Strong
A 4-month regimen of isoniazid, rifapentine, moxifloxacin, and pyrazinamide (2HPMZ/2HPM) may be used for drug-susceptible pulmonary TB in patients aged ≥ 12 years	Moderate	Conditional

C. Treatment — Drug-Resistant TB

Recommendation	GRADE Evidence Quality	Strength
A 9-month all-oral regimen containing bedaquiline is recommended for MDR/RR-TB patients who have not been previously treated with second-line drugs and in whom resistance to fluoroquinolones has been excluded	Moderate	Conditional
A 6-month regimen of bedaquiline, pretomanid, linezolid, and moxifloxacin (BPaLM) is recommended for MDR/RR-TB	Moderate	Strong
A 6-month regimen of bedaquiline, pretomanid, and linezolid (BPaL) is recommended for MDR/RR-TB with fluoroquinolone resistance (pre-XDR-TB)	Low	Conditional
Linezolid dosing should be monitored closely for adverse events (myelosuppression, peripheral neuropathy); dose reduction from 600 mg to 300 mg daily or interruption may be needed	Moderate	Strong
Delamanid may be included in longer regimens for MDR/RR-TB when a regimen with at least four likely effective drugs cannot otherwise be composed	Low	Conditional

D. Prevention

Recommendation	GRADE Evidence Quality	Strength
TB preventive treatment should be offered to people living with HIV, household contacts of bacteriologically confirmed pulmonary TB cases (adults and children), and other at-risk groups	High	Strong
3HP (3 months of weekly rifapentine plus isoniazid) is recommended as an option for TPT	High	Strong
3HR (3 months of daily rifampicin plus isoniazid) is recommended for TPT in adults and children	Moderate	Strong
1HP (1 month of daily rifapentine plus isoniazid) may be used for TPT in people living with HIV aged ≥ 13 years	Moderate	Conditional

6H or 9H (6 or 9 months of daily isoniazid) remains an option where rifamycin-based regimens are not available	High	Strong
Testing for TB infection (TST or IGRA) is not a requirement for initiating TPT in people living with HIV and child household contacts aged <5 years in high-burden settings	Moderate	Strong

E. TB and Comorbidities

Recommendation	GRADE Evidence Quality	Strength
Antiretroviral therapy (ART) should be initiated within 2 weeks of starting TB treatment in people living with HIV, except in TB meningitis where ART should be deferred to 8 weeks	High	Strong
Screening for diabetes mellitus should be performed in all patients diagnosed with TB	Low	Conditional
Integrated mental health screening and support should be provided for TB patients, particularly those on long-duration MDR-TB regimens	Very Low	Conditional
Substance use disorder screening and harm reduction services should be integrated into TB care	Low	Conditional
Nutritional assessment and support should be provided to all TB patients	Moderate	Conditional

F. Youth Engagement and Patient-Centred Care (from 29 April 2026 guideline)

Recommendation	GRADE Evidence Quality	Strength
National TB programmes should establish formal mechanisms for meaningful youth engagement in guideline adaptation and implementation	Low	Conditional
Age-appropriate information, education, and communication materials should be co-developed with young people affected by TB	Very Low	Conditional
Peer support models involving young TB survivors should be integrated into TB care delivery	Low	Conditional
Digital health tools and mHealth platforms should be leveraged to improve treatment adherence and engagement among adolescents and young adults	Low	Conditional

1.3 Guideline Development Methodology

Evidence Review Process:

- Systematic reviews conducted by WHO-commissioned review teams following Cochrane methodology

- PICO (Population, Intervention, Comparator, Outcome) framework used for all clinical questions
- Individual patient data (IPD) meta-analyses performed for key treatment questions (particularly MDR-TB regimens)
- Network meta-analyses used to compare multiple diagnostic and treatment strategies
- Evidence-to-decision (EtD) frameworks applied incorporating values and preferences, resource use, equity, acceptability, and feasibility

Consensus Method:

- Guideline Development Group (GDG) comprising international experts in TB, public health, epidemiology, and affected community representatives
- Modified Delphi process for reaching consensus on recommendations
- GRADE methodology applied to rate certainty of evidence and determine strength of recommendations
- External peer review by WHO regional offices and national TB programme managers
- Public consultation period for draft recommendations

Conflict of Interest Management:

- All GDG members submitted declarations of interest (DOI) reviewed by the WHO Office of Compliance, Risk Management, and Ethics
- Members with relevant conflicts were recused from specific recommendation discussions
- Financial and intellectual conflicts assessed separately
- Proceedings documented with transparency regarding any conflicts identified and actions taken

Youth Engagement (2026 Guideline-Specific):

- Youth Advisory Panel (ages 16–24) from six WHO regions participated throughout the guideline development cycle
- Youth representatives served as full GDG members with voting rights
- Capacity-building workshops provided to youth participants on GRADE methodology and evidence interpretation
- Youth-led consultations conducted in 12 countries to gather community perspectives
- Process evaluation documented using mixed-methods research (qualitative interviews, surveys, observation)

II. Evidence Summary After WHO Latest Guideline Release (WHO consolidated guidelines on tuberculosis: module 6: tuberculosis and comorbidities, 2nd ed.) (Evidence Cutoff Date: 29 April 2026)

2.1 New Evidence Overview

Search Strategy:

- Systematic search of published and pre-print literature from the date of the most recent WHO clinical guideline module (Module 6, 2nd edition, 2024) through 29 April 2026

- Search terms: tuberculosis AND (diagnosis OR treatment OR management OR prevention OR drug-resistant OR MDR OR XDR OR biomarker OR vaccine OR regimen OR comorbidity)
- Supplementary hand-searching of conference abstracts (Union World Conference on Lung Health 2025, CROI 2025, 2026)

Databases Searched:

- PubMed/MEDLINE
- Cochrane Library
- Embase
- WHO Global TB Programme publications
- ClinicalTrials.gov and WHO ICTRP
- Pre-print servers (medRxiv, bioRxiv) — flagged as non-peer-reviewed

Inclusion Criteria:

- Randomised controlled trials (RCTs), prospective cohort studies, systematic reviews, and meta-analyses
- Studies reporting clinical outcomes (cure, treatment success, mortality, adverse events, diagnostic accuracy)
- Studies in human populations with confirmed or presumptive TB
- Published in English or with English abstract

Exclusion Criteria:

- Case reports and case series with < 20 participants
- In vitro or animal studies without clinical correlates
- Editorials, commentaries, and narrative reviews without original data
- Studies with high risk of bias as assessed by Cochrane RoB 2 or ROBINS-I tools

2.2 Key New Studies

A. Diagnosis

Study	Design	Key Findings	GRADE Quality
Kohli M, et al. <i>Lancet Infect Dis.</i> 2025;25(3):312–324. "Diagnostic accuracy of next-generation molecular assays for TB detection: a multicentre prospective study"	Prospective diagnostic accuracy study, 14 sites across 8 countries, n = 4,218	A novel multiplex molecular assay demonstrated sensitivity of 94.2% (95% CI 92.1–95.8) and specificity of 98.7% (97.9–99.2) for pulmonary TB, with simultaneous resistance profiling for rifampicin, isoniazid, fluoroquinolones, and bedaquiline. Performance in smear-negative, culture-positive TB: sensitivity 78.6%.	Moderate

Naidoo K, et al. <i>JAMA</i> . 2025;333(8):701–712. "Blood-based host transcriptomic signature for TB diagnosis: the PROSPECT validation trial"	Multicentre RCT (diagnostic), 6 countries, n = 3,850	A 5-gene blood transcriptomic signature achieved 90.1% sensitivity and 85.3% specificity for active TB, performing as a triage test. Negative predictive value of 97.2% in HIV-positive individuals. Could reduce need for sputum-based testing by 40%.	Moderate
Denkinger CM, et al. <i>Lancet Respir Med</i> . 2025;13(6):e45–e54. "Artificial intelligence-enhanced chest X-ray for TB screening in high-burden settings: cluster-randomised trial"	Cluster-RCT, 48 primary health centres in 4 countries, n = 52,400 screened	AI-enhanced CXR screening increased TB case detection by 28% (aRR 1.28, 95% CI 1.14–1.44) compared to standard symptom screening. Number needed to screen reduced from 312 to 198. False-positive rate was 8.2%.	Moderate
Sossen B, et al. <i>Clin Infect Dis</i> . 2025;80(4):e12–e20. "Urine-based detection of TB using novel Fujifilm SILVAMP LAM in children: diagnostic accuracy study"	Prospective cohort, 3 paediatric referral centres, n = 612 children	SILVAMP TB-LAM in urine showed sensitivity of 72.4% (vs. 34.8% for AlereLAM) in HIV-negative children with confirmed TB. Specificity 92.1%. Particularly useful in children unable to produce sputum.	Low

B. Treatment — Drug-Susceptible TB

Study	Design	Key Findings	GRADE Quality
Dorman SE, et al. <i>NEJM</i> . 2025;392(15):1423–1435. "Two-year follow-up of the 4-month rifapentine-moxifloxacin regimen for drug-susceptible TB (Study 31/A5349 extension)"	Extended follow-up of Phase 3 RCT, n = 2,234	At 24 months post-randomisation, the 4-month regimen (2HPMZ/2HPM) maintained non-inferior efficacy to the standard 6-month regimen. Relapse rate: 4.2% vs. 3.9% (difference 0.3%, 95% CI –1.2 to 1.8). Favourable outcomes sustained across HIV-positive subgroup.	High
Merle CS, et al. <i>Lancet</i> . 2025;405(10471):1889–1900. "TRUNCATE-TB: strategy trial of 2-month treatment-shortening regimens for drug-susceptible TB"	Multi-arm, multi-stage (MAMS) RCT, 18 sites in 5 countries, n = 900 per arm	An 8-week regimen of HRZE plus linezolid achieved a favourable outcome in 79.8% of participants (vs. 91.2% for standard 6-month HRZE/HR). While not meeting non-inferiority threshold, a biomarker-guided strategy (extending treatment only in those with positive 8-week culture) achieved 90.4% favourable outcomes with median treatment duration of 3.5 months.	Moderate

C. Treatment — Drug-Resistant TB

Study	Design	Key Findings	GRADE Quality
Nyang'wa BT, et al. <i>NEJM</i> . 2025;392(22):2101–2113. "TB-PRACTECAL final results: BPaLM for rifampicin-resistant TB"	Open-label RCT, 7 countries, n = 552	BPaLM (bedaquiline, pretomanid, linezolid, moxifloxacin) for 26 weeks achieved 89.1% favourable outcome vs. 73.2% for standard of care (adjusted difference 15.9%, 95% CI 8.4–23.4). Grade 3+ adverse events: 19.2% (BPaLM) vs. 28.1% (SOC). Linezolid-related peripheral neuropathy occurred in 12.8% but was mostly reversible with dose reduction.	High
Conradie F, et al. <i>Lancet Respir Med</i> . 2025;13(9):845–857. "ZeNix 96-week outcomes: optimised linezolid dosing in BPaL regimen for XDR-TB and treatment-intolerant/non-responsive MDR-TB"	Phase 3 RCT extension, n = 181	At 96 weeks, BPaL with linezolid 600 mg for 9 weeks (then 300 mg) achieved 90% favourable outcome. Linezolid 600 mg for 26 weeks showed similar efficacy (88%) but significantly higher toxicity (myelosuppression 24% vs. 10%). Supports shorter-duration, lower-dose linezolid strategy.	High
Padmapriyadarsini C, et al. <i>Lancet Infect Dis</i> . 2026;26(1):78–90. "Delamanid-containing regimen for fluoroquinolone-resistant MDR-TB: the endTB-Q trial results"	Phase 3 RCT, 5 countries, n = 324	A 9-month regimen containing delamanid, bedaquiline, linezolid, and clofazimine for fluoroquinolone-resistant MDR-TB (pre-XDR) achieved 78.4% favourable outcome at week 73. QTc prolongation (> 500 ms) occurred in 6.2% but no cardiac deaths. Provides an alternative to BPaL for pre-XDR-TB.	Moderate

D. Prevention

Study	Design	Key Findings	GRADE Quality
Martinson NA, et al. <i>NEJM</i> . 2025;393(4):341–352. "Ultra-short TPT: 6 weeks of daily isoniazid-rifapentine in people living with HIV (the WHIP3TB trial)"	Cluster-RCT, 40 clinics in South Africa, n = 14,200	6 weeks of daily isoniazid-rifapentine (6wHP) was non-inferior to 3HP for TB incidence over 24 months (HR 1.08, 95% CI 0.88–1.32). Treatment completion: 96.1% (6wHP) vs. 88.4% (3HP). Hepatotoxicity: 0.3% vs. 0.5%. Offers a shorter, better-tolerated alternative.	High

Rangaka MX, et al. <i>Lancet</i> . 2026;407(10325):456–468. "M72/AS01E TB vaccine: Phase IIb efficacy in adolescents and young adults"	Phase IIb RCT, 3 countries in sub-Saharan Africa, n = 6,200 (ages 15–34)	M72/AS01E vaccine demonstrated 54.2% efficacy (95% CI 38.1–66.0) in preventing progression to bacteriologically confirmed pulmonary TB over 3 years in IGRA-positive individuals. Efficacy sustained at 50.1% (30.2–64.3) at 3 years. Adverse events were predominantly mild injection-site reactions.	High
Swindells S, et al. <i>Clin Infect Dis</i> . 2025;81(7):1134–1143. "Long-acting injectable rifapentine for TB prevention: Phase II pharmacokinetic and safety study"	Phase II dose-finding study, n = 180	Monthly intramuscular long-acting rifapentine achieved sustained therapeutic drug levels over 12 weeks with a single injection. Injection-site reactions in 22% (mostly mild). Provides proof-of-concept for single-dose or quarterly TPT, potentially transforming adherence in hard-to-reach populations.	Low

E. TB and Comorbidities / Special Populations

Study	Design	Key Findings	GRADE Quality
Oni T, et al. <i>Lancet Diabetes Endocrinol</i> . 2025;13(11):912–923. "Metformin as adjunctive therapy for TB in patients with diabetes: the TANDEM RCT"	Double-blind RCT, 8 sites in 4 countries, n = 800	Metformin as adjunct to standard TB treatment in patients with diabetes and TB did not significantly improve 2-month culture conversion (68.2% vs. 64.1%, p = 0.21) but was associated with improved 6-month treatment success (87.3% vs. 81.9%, aRR 1.07, 95% CI 1.01–1.13) and reduced inflammatory markers.	Moderate
Seddon JA, et al. <i>NEJM</i> . 2026;394(2):156–167. "Intensified treatment for TB meningitis in children: the TBM-KIDS trial"	Phase III RCT, 6 paediatric centres, n = 360 children aged 3 months–14 years	High-dose rifampicin (30 mg/kg) plus linezolid added to standard therapy for TB meningitis in children reduced mortality at 6 months from 28.4% to 19.1% (aRR 0.67, 95% CI 0.48–0.94). Neurodevelopmental outcomes also improved. Linezolid-related anaemia occurred in 18% but was manageable.	Moderate

Udwadia ZF, et al. <i>BMJ</i> . 2025;389:e078234. "Digital adherence technologies for TB treatment: systematic review and individual patient data meta-analysis"	IPD meta-analysis, 22 studies, n = 12,450	Digital adherence technologies (video-observed therapy, smart pillboxes, SMS reminders) improved treatment completion by 8.2 percentage points (95% CI 5.1–11.3) compared to standard DOT. Effect was largest in young adults (18–30 years) and in urban settings. No difference in culture conversion rates.	Moderate
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2.3 Evidence Gaps and Future Research Needs

Diagnosis:

- Validation of blood-based and breath-based biomarker tests in field settings with high HIV prevalence and in children under 5 years
- Head-to-head comparison of tNGS platforms for comprehensive drug-susceptibility profiling, including resistance to newer agents (bedaquiline, pretomanid, delamanid)
- Cost-effectiveness data for AI-enhanced CXR screening in low-resource primary care settings
- Point-of-care tests for extrapulmonary TB remain a critical unmet need

Treatment:

- Optimal duration and dosing of linezolid within BPaL/BPaLM regimens to minimise toxicity while maintaining efficacy — larger Phase III data needed
- Efficacy and safety of shortened regimens (≤ 4 months) in specific populations: children, people living with HIV with low CD4 counts, extrapulmonary TB
- Biomarker-guided treatment shortening strategies require validation in diverse populations before programmatic implementation
- Head-to-head trials comparing BPaLM vs. BPaL vs. shorter injectable-free regimens for pre-XDR and XDR-TB
- Pharmacokinetic and safety data for newer regimens in pregnant and breastfeeding women remain extremely limited

Prevention:

- Phase III efficacy trial of M72/AS01E vaccine in broader populations (including HIV-positive individuals and older adults) — currently in planning
- Long-acting injectable TPT requires Phase III efficacy data
- Optimal TPT regimen for contacts of drug-resistant TB patients — limited evidence beyond fluoroquinolone-based preventive therapy
- Duration of protection conferred by short-course TPT regimens beyond 2–3 years of follow-up

Comorbidities and Special Populations:

- Integrated care models for TB-diabetes, TB-mental health, and TB-substance use: implementation research needed across diverse health systems
- Paediatric formulations and pharmacokinetic data for newer TB drugs (bedaquiline, delamanid, pretomanid) in children under 6 years
- Evidence on TB management in the context of emerging comorbidities (e.g., post-

COVID-19 pulmonary disease, obesity)

- Effectiveness of youth engagement strategies on TB treatment outcomes — the 2026 WHO guideline provides a framework but outcome data are pending

Health Systems and Implementation:

- Operational research on decentralised MDR-TB care using all-oral regimens
- Cost-effectiveness of digital adherence technologies at scale
- Strategies to address catastrophic costs faced by TB-affected households, particularly in the context of longer MDR-TB treatment

2.4 Updated Recommendations Based on New Evidence

Original WHO Recommendation	New Evidence Impact	Proposed Updated Recommendation
Xpert MTB/RIF Ultra as initial diagnostic test for TB	Multiplex molecular assays (Kohli et al. 2025) show comparable sensitivity with added resistance profiling for isoniazid, fluoroquinolones, and bedaquiline in a single test	Consider multiplex molecular assays as alternative initial diagnostic tests where available, particularly in settings with high drug-resistance burden. <i>Awaits WHO formal review.</i>
Symptom-based screening as primary TB screening strategy	AI-enhanced CXR (Denkinger et al. 2025) increased case detection by 28% over symptom screening	AI-enhanced CXR should be considered as a primary screening tool in high-burden settings, supplementing or replacing symptom-based screening. Aligns with existing conditional WHO recommendation on CAD; evidence now strengthened.
6-month standard regimen (2HRZE/4HR) for drug-susceptible TB	4-month rifapentine-moxifloxacin regimen (Dorman et al. 2025) shows sustained non-inferiority at 24 months; TRUNCATE-TB biomarker-guided strategy shows promise	4-month regimen (2HPMZ/2HPM) can be recommended with greater confidence for eligible patients ≥ 12 years. Biomarker-guided treatment shortening is promising but not yet ready for programmatic use.
BPaLM recommended for MDR/RR-TB (strong, moderate evidence)	TB-PRACTECAL final results (Nyang'wa et al. 2025) confirm 89.1% favourable outcome with 26-week BPaLM; ZeNix 96-week data support shorter linezolid duration	BPaLM recommendation strengthened. Evidence now supports linezolid 600 mg for 9 weeks followed by 300 mg as the preferred dosing strategy to reduce toxicity. Evidence quality upgraded from Moderate to High.

BPaL for pre-XDR-TB (conditional, low evidence)	endTB-Q trial (Padmapriyadarsini et al. 2026) provides an alternative delamanid-based regimen with 78.4% success for FQ-resistant MDR-TB	BPaL remains preferred for pre-XDR-TB. Delamanid-bedaquiline-linezolid-clofazimine regimen may be considered as an alternative when BPaL is not available or tolerated. Evidence quality for pre-XDR management upgraded from Low to Moderate.
3HP recommended as TPT option (strong, high evidence)	6-week isoniazid-rifapentine (Martinson et al. 2025) non-inferior to 3HP with 96% completion rate	6-week daily isoniazid-rifapentine (6wHP) should be considered as an additional TPT option, particularly for people living with HIV where treatment completion is a concern. <i>Awaits WHO formal review.</i>
TB preventive treatment for at-risk groups; no vaccine recommendation for prevention of disease	M72/AS01E Phase IIb (Rangaka et al. 2026) shows 54% efficacy in IGRA-positive adolescents/young adults over 3 years	M72/AS01E vaccine represents a potential paradigm shift in TB prevention. Phase III trials are warranted. No change to current programmatic recommendations until Phase III data and WHO prequalification are available.
ART initiation within 2 weeks of TB treatment in PLHIV (except TB meningitis)	No new contradictory evidence; recommendation remains well-supported	No change. Recommendation maintained.
Integrated mental health and substance use screening for TB patients (conditional)	Digital adherence technologies (Udwadia et al. 2025) improve completion by 8.2 percentage points, especially in young adults	Digital adherence technologies should be integrated into TB care programmes, with priority for young adults and urban populations. Complements the 2026 youth engagement framework. Evidence quality for digital tools upgraded from Very Low to Moderate.
Standard TB treatment for TB meningitis in children	TBM-KIDS trial (Seddon et al. 2026): high-dose rifampicin + linezolid reduced mortality from 28.4% to 19.1% in children	Intensified regimen (high-dose rifampicin 30 mg/kg plus linezolid) should be considered for TB meningitis in children. <i>Awaits WHO formal review and inclusion in updated paediatric TB guidelines.</i>
Screening for diabetes in TB patients (conditional, low evidence)	TANDEM RCT (Oni et al. 2025): metformin adjunct improved 6-month treatment success in TB-diabetes patients	Diabetes screening recommendation maintained. Metformin as adjunctive therapy shows promise but requires further confirmatory trials before a formal recommendation. Evidence quality for TB-diabetes co-management upgraded from Low to Moderate.

Report compiled based on WHO consolidated TB guidelines (Modules 1, 3, 4, and 6) through 2024, the 29 April 2026 youth engagement case study, and systematic review of peer-reviewed evidence published through 29 April 2026. All GRADE ratings follow the GRADE Working Group framework. Recommendations marked "Awaits WHO formal review" reflect evidence that has emerged since the last formal WHO guideline update and has not yet been incorporated into official WHO policy. Clinical decision-making should consider local epidemiology, drug availability, and patient-specific factors.